

3(a)	<i>any one from:</i> stimulate / enhance / AW, macrophage response ; stimulate / enhance / AW, B-lymphocyte / B-cell / humoral, response ; A described part of humoral response e.g. increase antibody production stimulate / enhance / AW, T-killer / T-cytotoxic, response ; AVP ; e.g. suppresses immune response (during recovery) role in inflammation ref chemotaxis	1
3(b)(i)	<i>any two from:</i> <i>Leydig cell / not T-helper lymphocyte, because:</i> much / many / high proportion of, smooth endoplasmic reticulum ; (smooth endoplasmic reticulum present) that functions to synthesise, cholesterol / testosterone / steroid hormones / lipids ; no / little, rough endoplasmic reticulum ; rough endoplasmic reticulum is for polypeptide / protein synthesis ;	2
3(b)(ii)	transmission electron micrograph ; <i>plus one from:</i> cell ultrastructure can be seen / high resolution ; A detail e.g. membranes of mitochondrion cristae nuclear, envelope / membranes endoplasmic reticulum I rough / smooth (very) thin section / not surface contours / AW ; I ref. to magnification	2

Question	Answer	Marks
3(c)(i)	<i>one correct mechanism stated plus one associated detail</i> (passive / simple) diffusion ; R facilitated diffusion / diffusion through (membrane) protein from high(er) to low(er) concentration / down a concentration gradient ; AVP ; e.g. via phospholipid bilayer / hydrophobic core / fatty acid tails / AW or endocytosis ; description / pinching in / vesicle formation ; AVP ; e.g. ATP / energy, required	2
3(c)(ii)	<i>any two from</i> hydrophilic / water soluble / polar ; cannot cross, hydrophobic core (of phospholipid bilayer) / AW ; too large (to pass through gaps in fatty acid tails) ; no (specific), transport / carrier, proteins ;	2